Library Management System: A Comprehensive Design

An integrated solution for efficient library operations — catalogue, circulation, user management, and reporting. This deck covers requirements, architecture, diagrams, estimation, schedule and testing for a robust LMS suitable for academic or public libraries.

LIBRARY LMS

SYSTEM DESIGN

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Problem Statement

The library faces manual cataloguing, slow circulation, inaccurate availability, and fragmented user records. Goals: automate circulation, centralise catalogue, enable search, manage acquisitions, and provide analytics while ensuring data integrity and user privacy.

Primary Users

Librarians, Members (students/public), Administrators

Key Constraints

Budget limits, legacy data migration, offline availability, security & privacy

Acceptance Criteria

Search <1s, 99% uptime, accurate fines calculation, role-based access

SRS Summary

- Functional: Catalogue CRUD, Issue/Return, Fine computation, Reservation, Notifications
- Non-functional: Security (RBAC), Performance, Backup, Scalability

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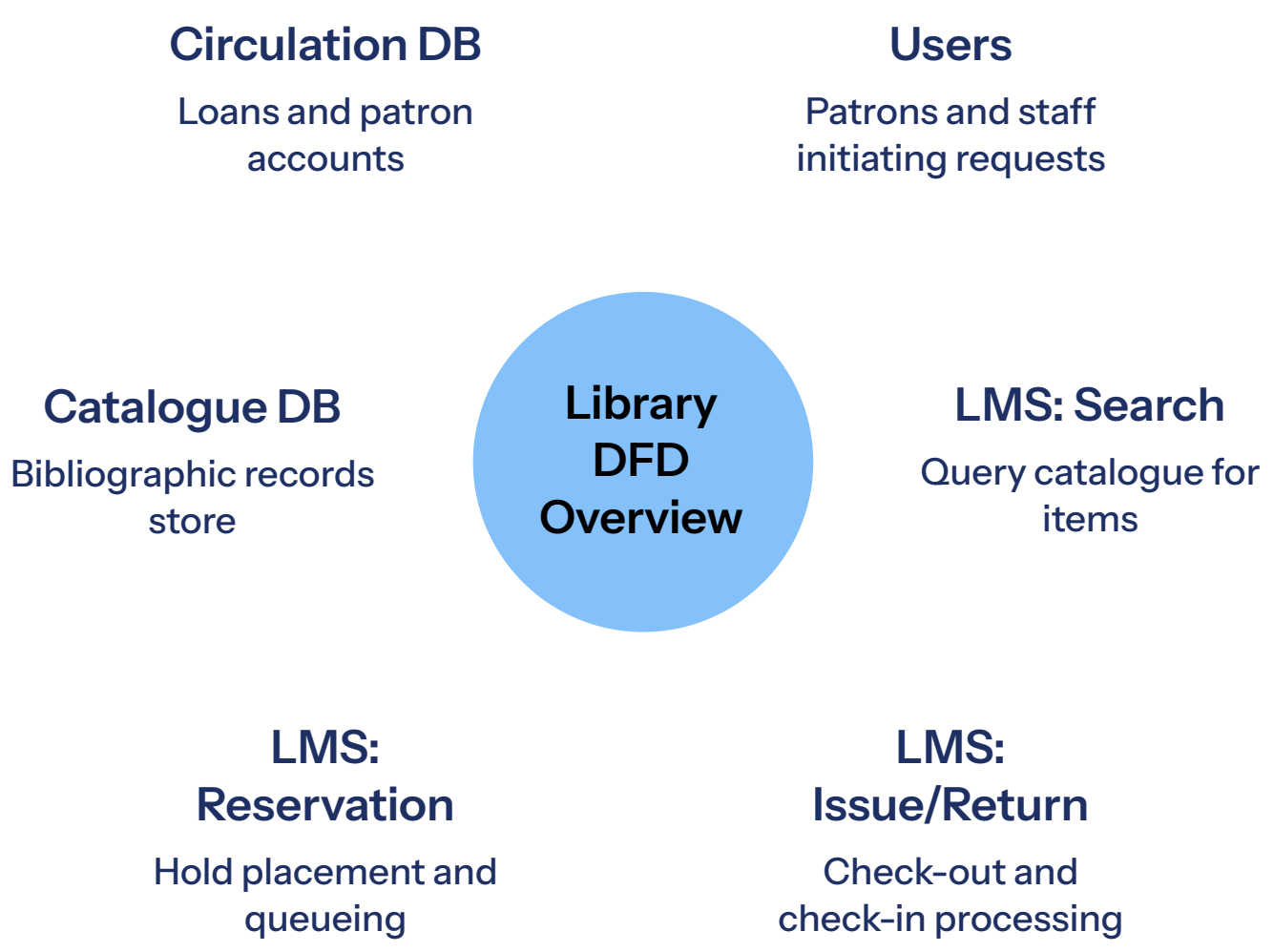
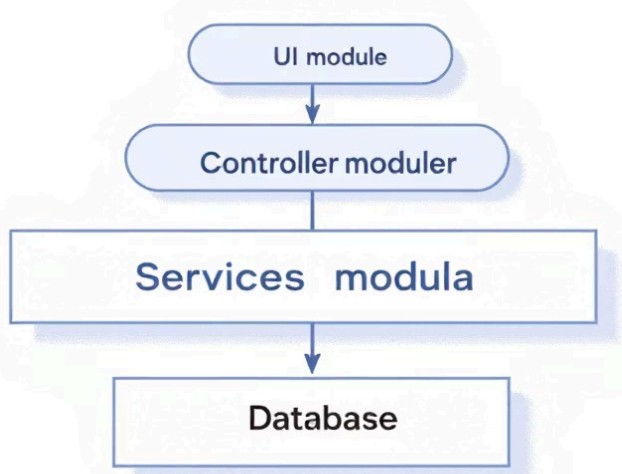
DFD — High Level (Level 0 & 1)

Top-level processes show data flow between Users, LMS, Catalogue, Circulation and External Services (Email/SMS). Level 1 expands issue/return, search, and reporting subprocesses.



Caption: The DFD clarifies how data moves: member requests trigger catalogue queries, circulation updates, and notifications. Use this to identify integration points and data stores.

Structured Chart



Structured chart shows module decomposition: Presentation → Controllers → Services (CatalogueService, CirculationService, UserService) → Persistence. Use for modular design & responsibility assignment.

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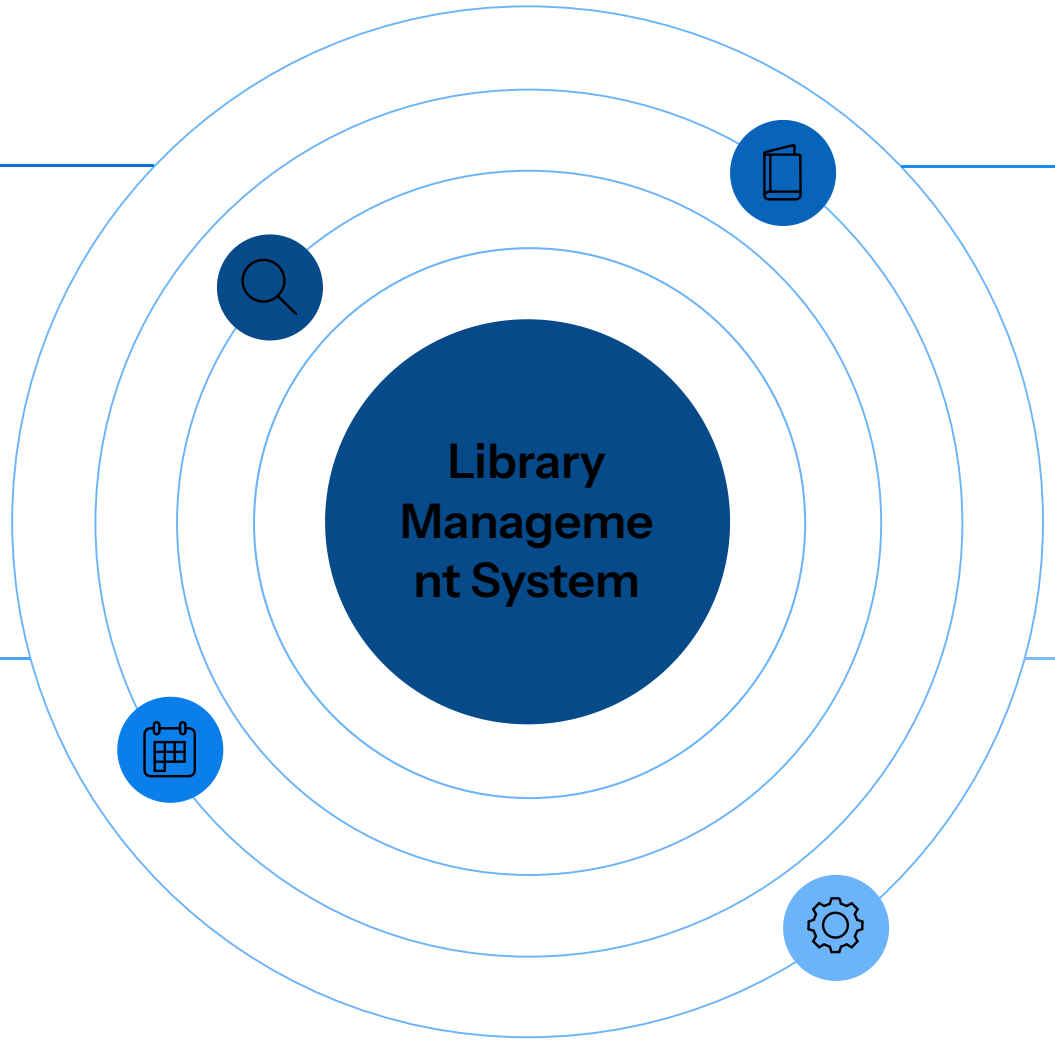
Use Case Diagram — Actors & Primary Flows



Search Catalogue

Find books by title, author, or ISBN

Issue & Return

Checkout and checkin books for members

Reserve & Pay Fine

Place holds and process fine payments

Manage & Report

Inventory management and report generation

Caption: Use cases capture user goals and system responsibilities. Prioritise Search, Issue/Return and Reservation for MVP; Admin functions for maintenance and reporting.

1

High Priority

Search, Issue, Return, User Authentication

2

Secondary

Reservation, Fines, Notifications

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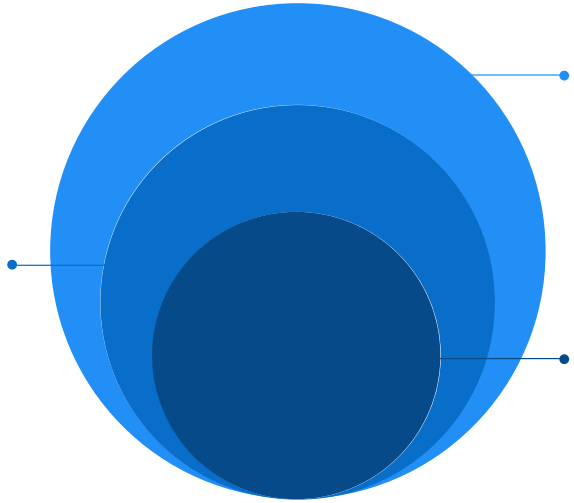
Class Diagram & Object Diagram

Class diagram outlines main entities and associations: Book, Copy, Member, Loan, Reservation, UserAccount, Fine, Inventory, Supplier. Emphasise composition between Book and Copy, aggregation between Member and Loan.



Interactions

Member → Loan (loanID, dueDate); Loan references Copy

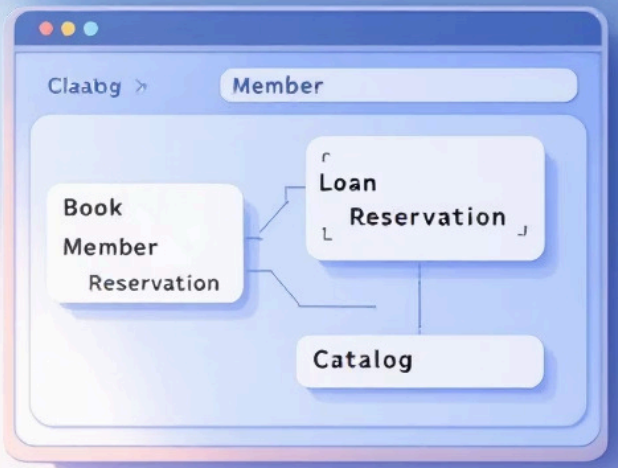
Contained Entities

Copy (copyID, status) — composition

Core Model

Book (ISBN, title)

Object Diagram (example snapshot): shows specific instances — Book: 'Data Structures', Copy: #23, Member: M102, Loan: L9001 (issued).



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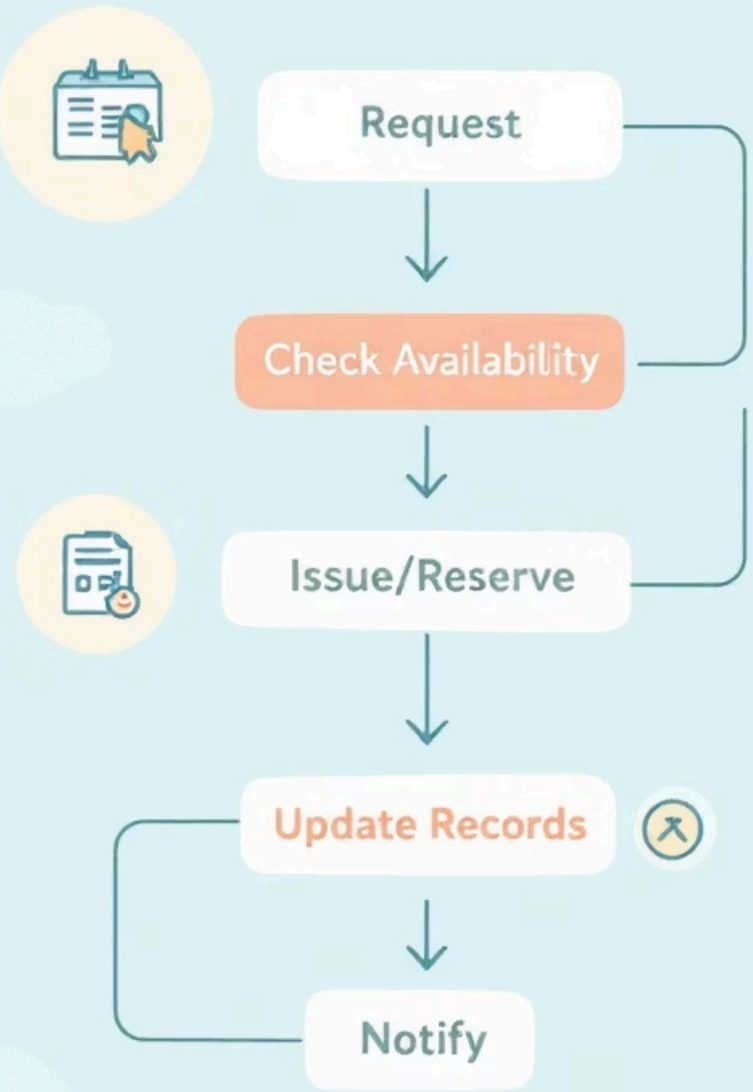
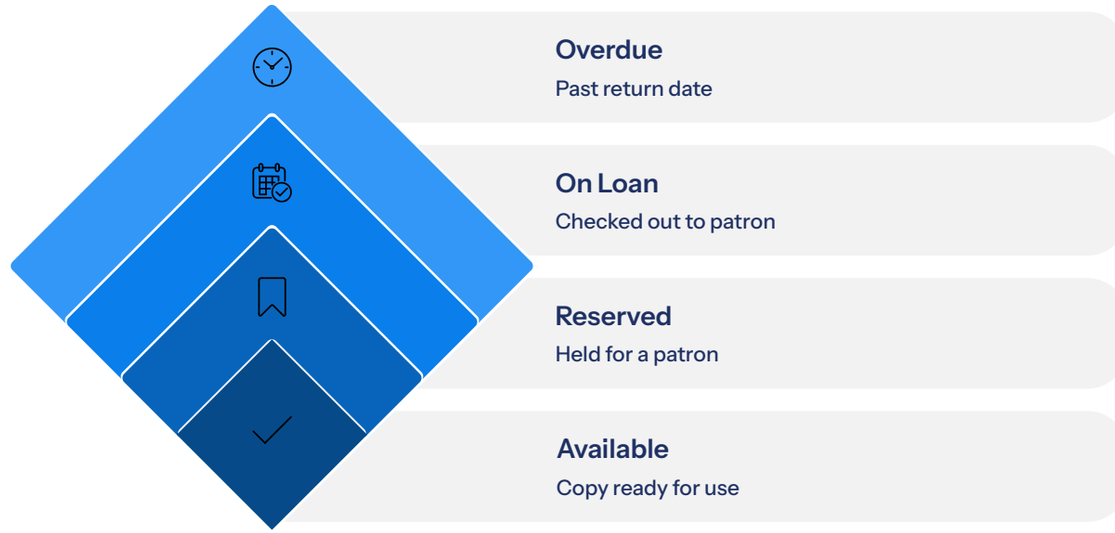
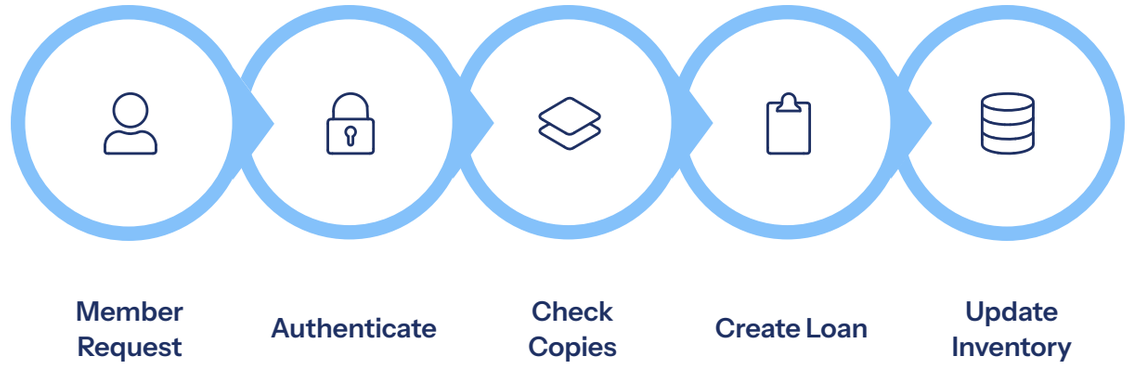


State-chart & Activity Diagrams



Caption: State-chart tracks lifecycle of a physical copy. Use it to drive UI states and validation rules.



Caption: Activity diagram emphasises parallel actions (update inventory and send notification) and decision points (available vs not available).

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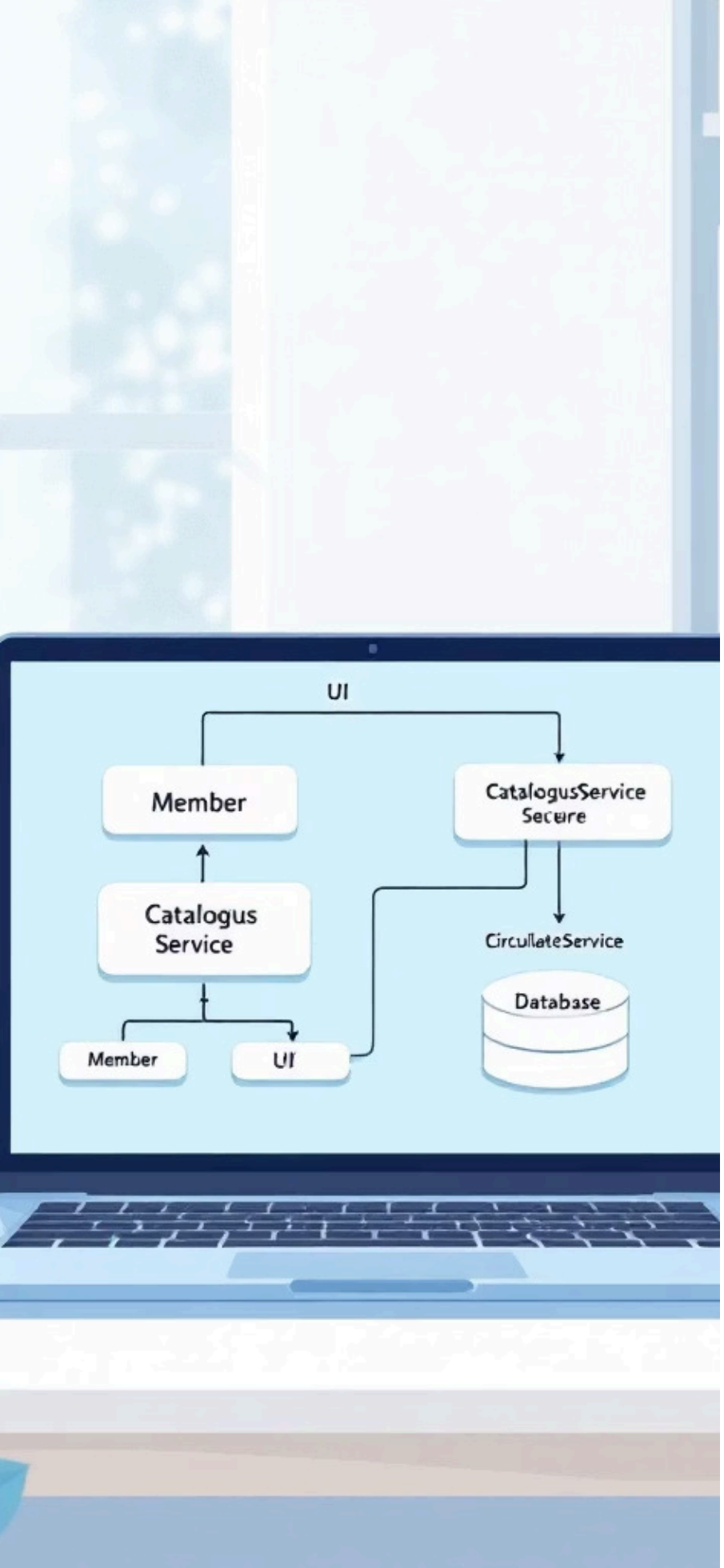
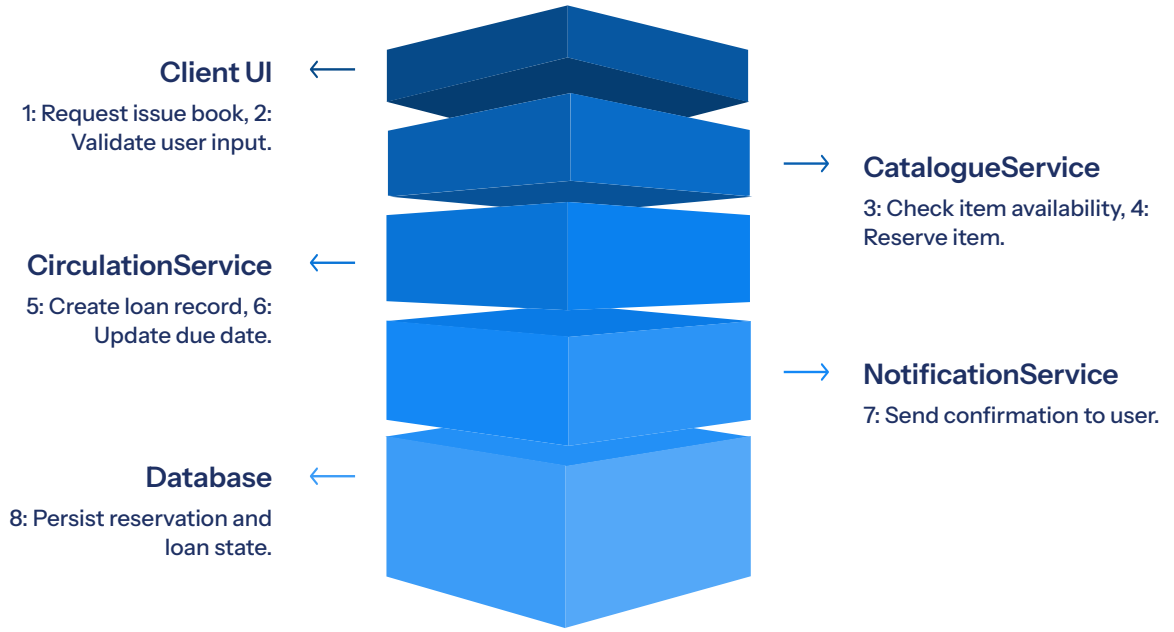


Sequence & Collaboration Diagrams

Caption: Sequence diagram details messages and timing between objects during issue flow. Useful for API design and transaction boundaries.



Caption: Collaboration view shows object links and responsibilities — valuable when designing component interactions and concurrency handling.

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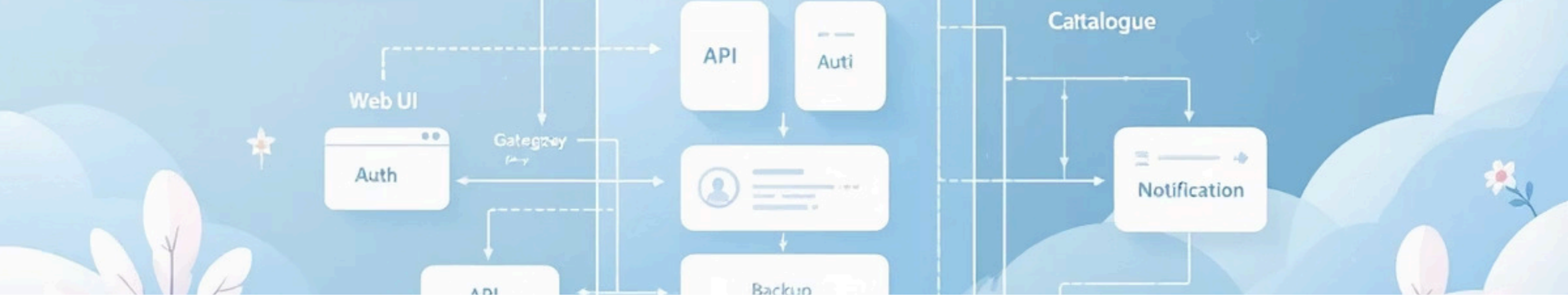
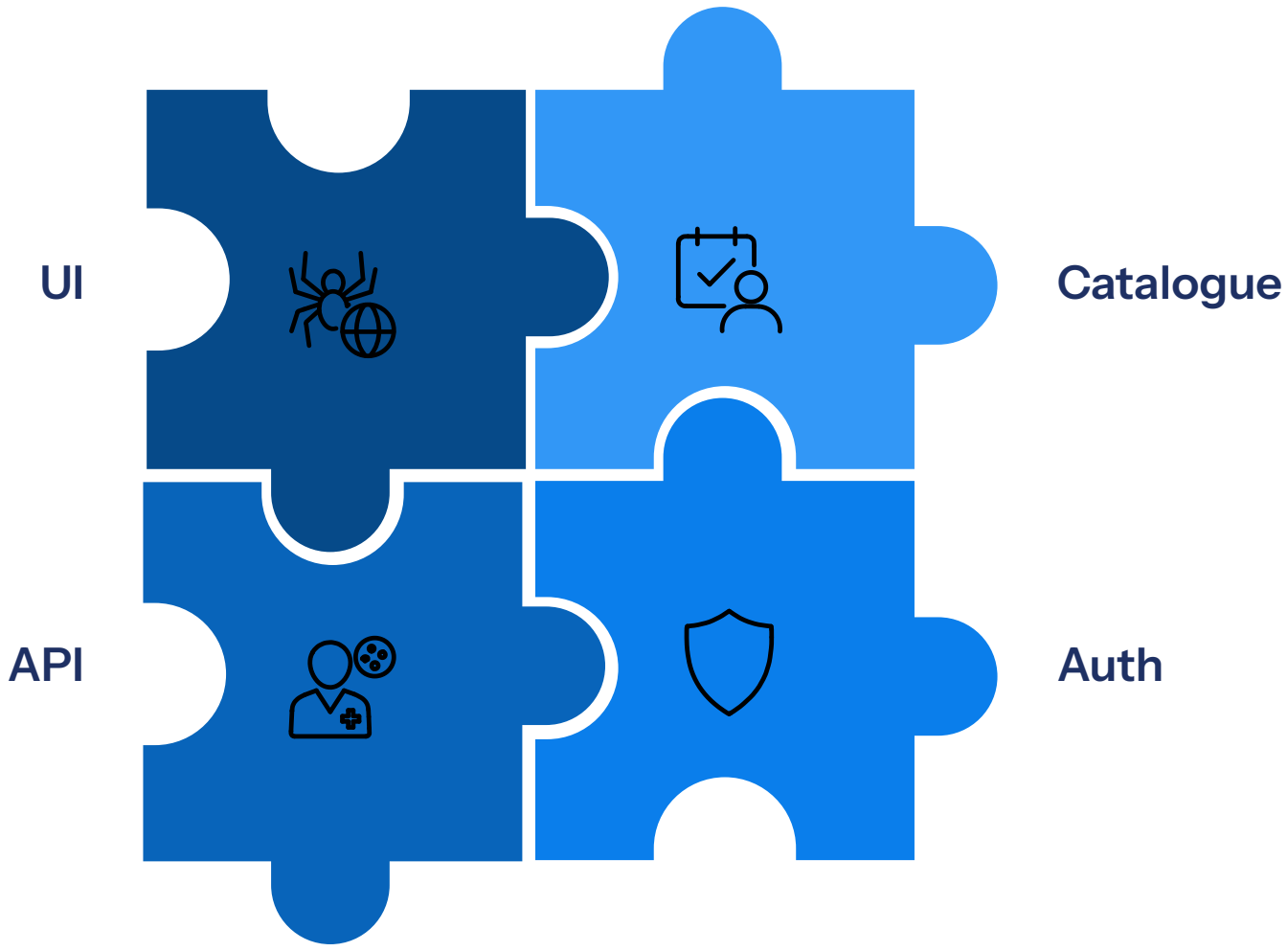
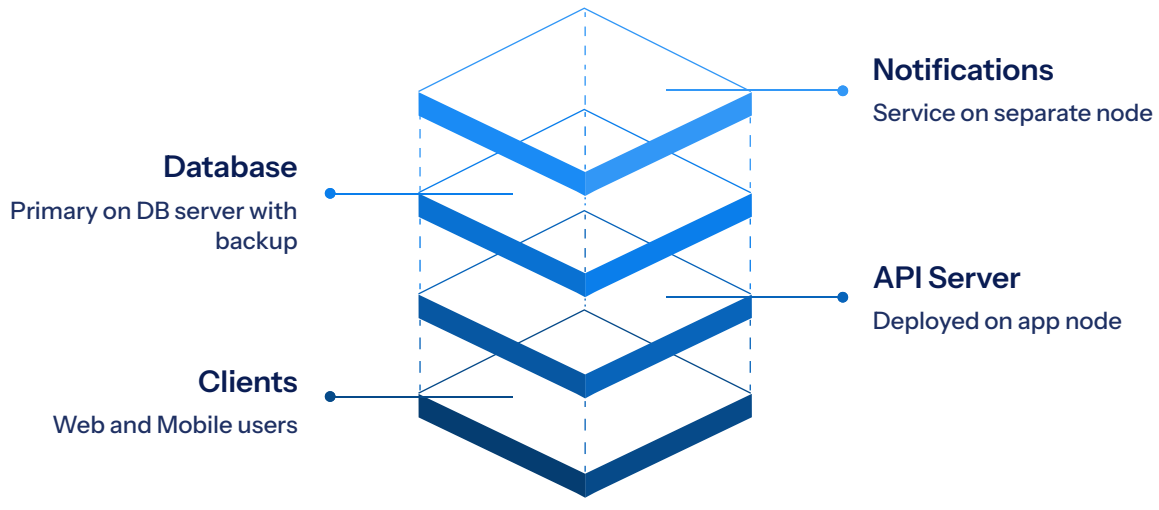


Component & Deployment Diagrams



Caption: Components define code boundaries for teams; design microservices around Catalogue and Circulation for scalability.



Deployment notes: Use separate DB for read replicas, enable secure communication (TLS), containerise services for portability, and include offline kiosk mode for intermittent networks.

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Project Estimation — Function Point (FP) Method

Brief FP steps: identify functional components (EI, EO, EQ, ILF, EIF), classify complexity, compute Unadjusted Function Points (UFP), apply Value Adjustment Factor (VAF) to get Adjusted FP, convert to effort using productivity factor.

1 Step 1: Identify

EI: Search, Issue; EO: Notifications; ILF: Catalogue, Member DB; EIF: External Payment Gateway

2 Step 2: Count

Example UFP ≈ 160 (sample counts). VAF = 0.85 → Adjusted FP ≈ 136

3 Step 3: Effort

Conversion: 15 FP/person-month → ~9 person-months for core system (adjust per organisation)

Use FP for early budgeting, then refine with LOC or story-point estimates during planning.

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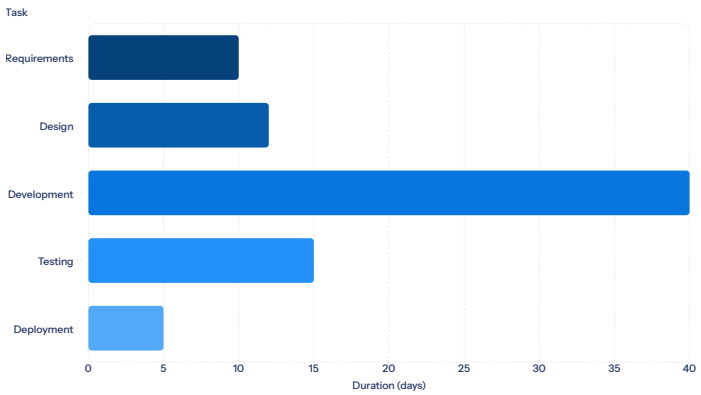
Schedule & Testing

Gantt / PERT Overview



PERT note: Identify critical path (Design → Development → Testing). Add 10% contingency for integration risks.

Test Cases Summary

Unit Tests

Validate CatalogueService methods, Loan calculation, fine computation. Edge cases: zero-availability, date boundary.

Integration Tests

Catalogue ↔️ Circulation ↔️ Notification integration; DB transaction rollback and concurrency tests.

White Box

Path coverage for fine calculation, authentication flows, and data validation routines.

Black Box

Functional acceptance: Search, Issue/Return, Reservation, UI workflows and accessibility checks.

Prioritise regression suites after each release. Target >=80% automated coverage on critical modules.

Next steps: review diagrams with stakeholders, finalise FP numbers with domain expert, begin iterative sprints with continuous testing and deployment pipelines.

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